

Lecture 05: Linear Algebra

Introduction to Machine Learning

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Except explicitly cited, the reference for the material in slides is:

- Murphy, K. P. (2022). *Probabilistic machine learning: an introduction*. MIT press.

Section 1

Basic Definitions

Basic Definitions

Vectors

In this course we assume column vectors represented by:

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = (x_1, x_2, \dots, x_n)$$

Matrices

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}$$

Matrix Rows

$$\mathbf{A} = \begin{bmatrix} -\mathbf{A}_{1,:}^T & - \\ -\mathbf{A}_{2,:}^T & - \\ \vdots & \\ -\mathbf{A}_{m,:}^T & - \end{bmatrix} = [\mathbf{A}_{1,:}^T \quad ; \quad \mathbf{A}_{2,:}^T \quad ; \quad \dots \quad ; \quad \mathbf{A}_{m,:}^T]$$

Matrix Columns

$$\mathbf{A} = \left[\begin{array}{c|c|c|c} | & | & & | \\ \mathbf{A}_{:,1} & \mathbf{A}_{:,2} & \dots & \mathbf{A}_{:,n} \\ | & | & & | \end{array} \right] = [\mathbf{A}_{:,1} \quad , \quad \mathbf{A}_{:,2} \quad , \quad \dots \quad , \quad \mathbf{A}_{:,n}]$$

Vectorizing Operator

$$\text{vec}(\mathbf{A}) = [\mathbf{A}_{:,1}; \dots; \mathbf{A}_{:,n}] \in \mathbb{R}^{mn \times 1}$$

I-vectorizing Operator

$$\mathbf{A} = \text{ivec}(\text{vec}(\mathbf{A}), \mathcal{O})$$

Section 2

Vector Space

Vector Space

Vector Space

A vector space is a set of vectors $\mathbf{x} \in \mathbb{R}^n$, denoted $\mathcal{V}$, such that:

- It is closed under vector addition: *if* $\mathbf{x}, \mathbf{y} \in \mathcal{V} \Rightarrow \mathbf{x} + \mathbf{y} \in \mathcal{V}$
- It is closed under multiplication by a real scalar $c \in \mathbb{R}$: *if* $\mathbf{x} \in \mathcal{V} \Rightarrow c\mathbf{x} \in \mathcal{V}$

Linear Independence

A set of vectors $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n\}$ is said to be (linearly) dependent if:

$$\exists j : \mathbf{x}_j = \sum_{i, i \neq j} \alpha_i \mathbf{x}_i$$

Otherwise the set is said to be (linearly) independent.

Span

The span of a set of vectors $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n\}$ is defined as:

$$\text{span}(\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n\}) \triangleq \left\{ \mathbf{v} : \mathbf{v} = \sum_{i=1}^n \alpha_i \mathbf{x}_i, \alpha_i \in \mathbb{R} \right\}$$

Section 3

Products

Matrix-Vector Product

Matrix-Vector Product

Assume $\mathbf{A} \in \mathbb{R}^{m \times n}$ and $\mathbf{x} \in \mathbb{R}^n$. Then the product vector $\mathbf{y} = \mathbf{Ax} \in \mathbb{R}^m$ can be viewed as follows:

View 1

$$\mathbf{y} = \mathbf{Ax} = \begin{bmatrix} - & \hat{\mathbf{a}}_1^T & - \\ - & \hat{\mathbf{a}}_2^T & - \\ & \vdots & \\ - & \hat{\mathbf{a}}_m^T & - \end{bmatrix} \mathbf{x} = \begin{bmatrix} \hat{\mathbf{a}}_1^T \mathbf{x} \\ \hat{\mathbf{a}}_2^T \mathbf{x} \\ \vdots \\ \hat{\mathbf{a}}_m^T \mathbf{x} \end{bmatrix}$$

View 2

$$\mathbf{y} = \mathbf{Ax} = \begin{bmatrix} | & | & & | \\ \mathbf{a}_1 & \mathbf{a}_2 & \dots & \mathbf{a}_n \\ | & | & & | \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} | \\ \mathbf{a}_1 \\ | \end{bmatrix} x_1 + \begin{bmatrix} | \\ \mathbf{a}_2 \\ | \end{bmatrix} x_2 + \dots + \begin{bmatrix} | \\ \mathbf{a}_n \\ | \end{bmatrix} x_n$$

Matrix-Matrix Product

Matrix-Matrix Product

Assume $\mathbf{A} \in \mathbb{R}^{m \times n}$ and $\mathbf{B} \in \mathbb{R}^{n \times p}$. Then the product vector $\mathbf{C} = \mathbf{AB} \in \mathbb{R}^{m \times p}$ can be viewed as follows:

View 1

$$\mathbf{C} = \mathbf{AB} = \begin{bmatrix} - & \hat{\mathbf{a}}_1^T & - \\ - & \hat{\mathbf{a}}_2^T & - \\ & \vdots & \\ - & \hat{\mathbf{a}}_m^T & - \end{bmatrix} \begin{bmatrix} | & | & \dots & | \\ \mathbf{b}_1 & \mathbf{b}_2 & \dots & \mathbf{b}_p \\ | & | & & | \end{bmatrix} = \begin{bmatrix} \hat{\mathbf{a}}_1^T \mathbf{b}_1 & \hat{\mathbf{a}}_1^T \mathbf{b}_2 & \dots & \hat{\mathbf{a}}_1^T \mathbf{b}_p \\ \hat{\mathbf{a}}_2^T \mathbf{b}_1 & \hat{\mathbf{a}}_2^T \mathbf{b}_2 & \dots & \hat{\mathbf{a}}_2^T \mathbf{b}_p \\ \vdots & \vdots & \ddots & \vdots \\ \hat{\mathbf{a}}_m^T \mathbf{b}_1 & \hat{\mathbf{a}}_m^T \mathbf{b}_2 & \dots & \hat{\mathbf{a}}_m^T \mathbf{b}_p \end{bmatrix}$$

View 2

$$\mathbf{C} = \mathbf{AB} = \begin{bmatrix} | & | & \dots & | \\ \mathbf{a}_1 & \mathbf{a}_2 & \dots & \mathbf{a}_n \\ | & | & & | \end{bmatrix} \begin{bmatrix} - & \hat{\mathbf{b}}_1^T & - \\ - & \hat{\mathbf{b}}_2^T & - \\ & \vdots & \\ - & \hat{\mathbf{b}}_n^T & - \end{bmatrix} = \sum_{i=1}^n \mathbf{a}_i \hat{\mathbf{b}}_i^T$$

Matrix-Matrix Product

Matrix-Matrix Product

Assume $\mathbf{A} \in \mathbb{R}^{m \times n}$ and $\mathbf{B} \in \mathbb{R}^{n \times p}$. Then the product vector $\mathbf{C} = \mathbf{AB} \in \mathbb{R}^{m \times p}$ can be viewed as follows:

View 3

$$\mathbf{C} = \mathbf{AB} = \mathbf{A} \begin{bmatrix} | & | & \dots & | \\ \mathbf{b}_1 & \mathbf{b}_2 & \dots & \mathbf{b}_p \\ | & | & & | \end{bmatrix} = \begin{bmatrix} | & | & \dots & | \\ \mathbf{Ab}_1 & \mathbf{Ab}_2 & \dots & \mathbf{Ab}_p \\ | & | & & | \end{bmatrix}$$

View 4

$$\mathbf{C} = \mathbf{AB} = \begin{bmatrix} - & \hat{\mathbf{a}}_1^T & - \\ - & \hat{\mathbf{a}}_2^T & - \\ & \vdots & \\ - & \hat{\mathbf{a}}_m^T & - \end{bmatrix} \mathbf{B} = \begin{bmatrix} - & \hat{\mathbf{a}}_1^T \mathbf{B} & - \\ - & \hat{\mathbf{a}}_2^T \mathbf{B} & - \\ & \vdots & \\ - & \hat{\mathbf{a}}_m^T \mathbf{B} & - \end{bmatrix}$$

Range and Null Spaces

Range of a Matrix

Assume $\mathbf{A} \in \mathbb{R}^{m \times n}$. The range or columns space of $\mathbf{A}$ is the span of the columns of $\mathbf{A}$ as:

$$\text{range}(\mathbf{A}) \triangleq \{\mathbf{v} \in \mathbb{R}^m : \mathbf{v} = \mathbf{A}\mathbf{x}, \mathbf{x} \in \mathbb{R}^n\}$$

Null Space of a Matrix

Assume $\mathbf{A} \in \mathbb{R}^{m \times n}$. The null space of $\mathbf{A}$ is the set of all vectors $\mathbf{x}$ that get mapped to the null vector when multiplied by $\mathbf{A}$ as:

$$\text{nullspace}(\mathbf{A}) \triangleq \{\mathbf{x} \in \mathbb{R}^n : \mathbf{A}\mathbf{x} = \mathbf{0}\}$$

Section 4

Norms

Definition

Norm is any function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ that satisfies the following properties:

- ① $\forall \mathbf{x} \in \mathbb{R}^n \Rightarrow f(\mathbf{x}) \geq 0$ (non-negativity)
- ② $f(\mathbf{x}) = 0$ iff $\mathbf{x} = 0$ (definiteness)
- ③ $\forall \mathbf{x} \in \mathbb{R}^n, \forall t \in \mathbb{R} \Rightarrow f(t\mathbf{x}) = |t|f(\mathbf{x})$ (absolute value homogeneity)
- ④ $\forall \mathbf{x} \in \mathbb{R}^n, \forall \mathbf{y} \in \mathbb{R}^n \Rightarrow f(\mathbf{x} + \mathbf{y}) \leq f(\mathbf{x}) + f(\mathbf{y})$ (triangle inequality)

Examples of Vector Norm

- p-norm (ℓ_p): $\|\mathbf{x}\|_p = (\sum_{i=1}^n |x_i|^p)^{1/p}, p \geq 1 \Rightarrow \begin{cases} \ell_1 : \|\mathbf{x}\|_1 = \sum_{i=1}^n |x_i| \\ \ell_2 : \|\mathbf{x}\|_2 = \sqrt{\sum_{i=1}^n x_i^2} \\ \ell_\infty : \|\mathbf{x}\|_\infty = \max_i |x_i| \end{cases}$
- 0-norm (ℓ_0): $\|\mathbf{x}\|_0 = \sum_{i=1}^n \mathbb{I}(|x_i| > 0)$ (Pseudo norm due to inhomogeneity)

Examples of Matrix Norm

Assume matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$, then:

- p-norm (ℓ_p): $\|\mathbf{A}\|_p = \max_{\mathbf{x} \neq 0} \frac{\|\mathbf{Ax}\|_p}{\|\mathbf{x}\|_p} = \max_{\|\mathbf{x}\|=1} \|\mathbf{Ax}\|_p$
- Frobenius norm (ℓ_F): $\|\mathbf{A}\|_F = \sqrt{\sum_{i=1}^m \sum_{j=1}^n a_{ij}^2} = \|\text{vec}(\mathbf{A})\|_2$

Section 5

Matrix Operators

Trace of a Square Matrix

Definition

The trace of a square matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$, denoted $\text{tr}(\mathbf{A})$, is the sum of diagonal elements in the matrix as:

$$\text{tr}(\mathbf{A}) \triangleq \sum_{i=1}^n A_{ii}$$

Properties

Assume matrices $\mathbf{A}, \mathbf{B} \in \mathbb{R}^{n \times n}$ and scalar $c \in \mathbb{R}$.

- $\text{tr}(\mathbf{A}) = \text{tr}(\mathbf{A}^T)$
- $\text{tr}(\mathbf{A} + \mathbf{B}) = \text{tr}(\mathbf{A}) + \text{tr}(\mathbf{B})$
- $\text{tr}(c\mathbf{A}) = c \text{tr}(\mathbf{A})$
- $\text{tr}(\mathbf{AB}) = \text{tr}(\mathbf{BA})$

Trace of a Square Matrix

Cyclic Permutation Property

For real matrices $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{C}$ where $\mathbf{ABC}$ is square, then we have:

$$\text{tr}(\mathbf{ABC}) = \text{tr}(\mathbf{CAB}) = \text{tr}(\mathbf{BCA})$$

Determinant of a Square Matrix

Minor

Assume $\mathbf{A} \in \mathbb{R}^{n \times n}$. The (i, j) minor, denoted $\mathbf{A}_{ij}$ is the matrix obtained from $\mathbf{A}$ by deleting the i -th row and the j -th column.

Cofactor

Assume $\mathbf{A} \in \mathbb{R}^{n \times n}$. The (i, j) cofactor, denoted C_{ij} is: $C_{ij} = (-1)^{i+j} \det(\mathbf{A}_{ij})$, where $\det(\mathbf{A}_{ij})$ is the determinant of (i, j) minor.

Determinant

The determinant of a square matrix, denoted $\det(\mathbf{A})$ or $|\mathbf{A}|$, is a measure of how much it changes a unit volume when viewed as a linear transformation and is defined as:

$$\det(\mathbf{A}) = \sum_{i=1}^n a_{i1} C_{i1}$$

Condition Number of a Square Matrix

Condition Number

The condition number of a square matrix $\mathbf{A}$ is a measure for the stability of linear equation set $\mathbf{Ax} = \mathbf{b}$ and is defined as follows: $\kappa(\mathbf{A}) \triangleq \|\mathbf{A}\| \times \|\mathbf{A}^{-1}\|$. A suitable option for the matrix norm is ℓ_2 norm which result in $\kappa(\mathbf{A}) \geq 1$.

Matrix Conditioning

Assume square matrix $\mathbf{A}$. Based on the condition number, this matrix can be divided into two categories:

- $\mathbf{A}$ is ill-conditioned if $\kappa(\mathbf{A})$ is large.
- $\mathbf{A}$ is well-conditioned if $\kappa(\mathbf{A})$ is small (close to 1).

Condition Number of a Square Matrix

Frame Title

In a linear system of equations $\mathbf{Ax} = \mathbf{b}$, assume we change $\mathbf{b}$ to $\mathbf{b} + \Delta\mathbf{b}$. Compute the change in $\mathbf{x}$ vector ($\Delta\mathbf{x}$) for the following two matrices:

- $\mathbf{A} = 0.1\mathbf{I}_{100 \times 100}$ ($\kappa(\mathbf{A}) = 1, \det(\mathbf{A}) = 10^{-100}$):

$$\Delta\mathbf{x} = \mathbf{A}^{-1}\Delta\mathbf{b} = 10\mathbf{I}\Delta\mathbf{b} = 10\Delta\mathbf{b}$$

- $\mathbf{A} = 0.5 \begin{bmatrix} 1 & 1 \\ 1 + 10^{-10} & 1 - 10^{-10} \end{bmatrix}$ ($\kappa(\mathbf{A}) = 2 \times 10^{10}, \det(\mathbf{A}) = -2 \times 10^{-10}$):

$$\Delta\mathbf{x} = \mathbf{A}^{-1}\Delta\mathbf{b} = \frac{1}{-2 \times 10^{-10}} \begin{bmatrix} \Delta b_1 - 10^{10}(\Delta b_1 - \Delta b_2) \\ \Delta b_2 + 10^{10}(\Delta b_1 - \Delta b_2) \end{bmatrix}$$

Section 6

Special Matrices

Diagonal Matrix

- Diagonal matrix:

$$\mathbf{D} = \begin{bmatrix} d_1 & & & \\ & d_2 & & \\ & & \ddots & \\ & & & d_n \end{bmatrix} = \text{diag}(d_1, d_2, \dots, d_n)$$

- Block diagonal Matrix: A square matrix with square matrices in the main diagonal blocks and zero matrices in all off-diagonal blocks as:

$$\mathbf{A} = \begin{bmatrix} \mathbf{A}_1 & & & \\ & \mathbf{A}_2 & & \\ & & \ddots & \\ & & & \mathbf{A}_n \end{bmatrix} = \text{diag}(\mathbf{A}_1, \mathbf{A}_2, \dots, \mathbf{A}_n)$$

Band-diagonal Matrix

A band-diagonal matrix only has non-zero entries along the diagonal, and on k sides of the diagonal (k is known as bandwidth).

Tridiagonal Matrix

Tridiagonal matrix is a band-diagonal matrix with $k = 1$. A sample 6×6 tridiagonal matrix is:

$$A = \begin{bmatrix} a_{11} & a_{12} & 0 & 0 & 0 & 0 \\ a_{21} & a_{22} & a_{23} & 0 & 0 & 0 \\ 0 & a_{32} & a_{33} & a_{34} & 0 & 0 \\ 0 & 0 & a_{43} & a_{44} & a_{45} & 0 \\ 0 & 0 & 0 & a_{54} & a_{55} & a_{56} \\ 0 & 0 & 0 & 0 & a_{65} & a_{66} \end{bmatrix}$$

Triangular Matrix

Lower Triangular Matrix

$$L = \begin{bmatrix} l_{11} & & & & \\ l_{21} & l_{22} & & & \\ l_{31} & l_{32} & l_{33} & & \\ \vdots & \vdots & \ddots & \ddots & \\ l_{n1} & l_{n2} & \dots & l_{n(n-1)} & l_{nn} \end{bmatrix}$$

Upper Triangular Matrix

$$U = \begin{bmatrix} u_{11} & u_{12} & u_{13} & \dots & u_{1n} \\ & u_{22} & u_{23} & \dots & u_{2n} \\ & & \ddots & \ddots & \vdots \\ & & & \ddots & u_{(n-1)n} \\ & & & & u_{nn} \end{bmatrix}$$

Definite and Indefinite Matrices

Symmetric Matrix

Matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ is symmetric iff $\mathbf{A} = \mathbf{A}^T$ (We usually show this by $\mathbf{A} \in \mathbb{S}^n$)

Definite and Indefinite Matrices

Suppose $\mathbf{A} \in \mathbb{S}^n$ and arbitrary nonzero vector $\mathbf{v} \in \mathbb{R}^n \setminus \{0\}$ then:

- $\mathbf{A}$ is positive definite (PD), denoted $\mathbf{A} \succ 0$, iff: $\mathbf{v}^T \mathbf{A} \mathbf{v} > 0$
- $\mathbf{A}$ is positive semidefinite (PSD), denoted $\mathbf{A} \succeq 0$, iff: $\mathbf{v}^T \mathbf{A} \mathbf{v} \geq 0$
- $\mathbf{A}$ is negative definite (ND), denoted $\mathbf{A} \prec 0$, iff: $\mathbf{v}^T \mathbf{A} \mathbf{v} < 0$
- $\mathbf{A}$ is negative semidefinite (NSD), denoted $\mathbf{A} \preceq 0$, iff: $\mathbf{v}^T \mathbf{A} \mathbf{v} \leq 0$
- $\mathbf{A}$ is indefinite iff it is none of the above.

Orthogonal Square Matrices

Orthogonal Square Matrices

$\mathbf{A} = \begin{bmatrix} | & | & & | \\ \mathbf{a}_1 & \mathbf{a}_2 & \dots & \mathbf{a}_n \\ | & | & & | \end{bmatrix}$ is orthogonal iff:

$$\mathbf{a}_i^T \mathbf{a}_j = \begin{cases} 0 & \text{if } i \neq j \\ 1 & \text{if } i = j \end{cases}$$

Section 7

Inverse Matrix

Inverse Matrix

The inverse of a square matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$, denoted $\mathbf{A}^{-1}$, is the unique matrix such that:

$$\mathbf{A}^{-1}\mathbf{A} = \mathbf{A}\mathbf{A}^{-1} = \mathbf{I}$$

Singular Matrix

$\mathbf{A}^{-1}$ exists iff $\det(\mathbf{A}) \neq 0$. If $\det(\mathbf{A}) = 0$, $\mathbf{A}$ is called a singular matrix.

Section 8

Eigenvalue Decomposition

Eigenvalue and Eigenvector

Eigenvalue and Eigenvector

Assume a square matrix $\mathbf{A} \in \mathbb{R}^{2 \times 2}$, we say that $\lambda \in \mathbb{R}$ is an eigenvalue of $\mathbf{A}$ and $\mathbf{u} \in \mathbb{R}^n$ is the corresponding eigenvector if:

$$\mathbf{A}\mathbf{u} = \lambda\mathbf{u}, \mathbf{u} \neq \mathbf{0}$$

“The” Eigenvector

For any eigenvector $\mathbf{u} \in \mathbb{R}^n \setminus \{\mathbf{0}\}$ and scalar $c \in \mathbb{R} \setminus \{0\}$, $c\mathbf{u}$ is also an eigenvector. “The” eigenvector is normalized to have unit length.

Eigenvalue and Eigenvector

Characteristic Equation

$(\lambda, \mathbf{u})$ is (eigenvalue, eigenvector) pair if:

$$(\lambda \mathbf{I} - \mathbf{A})\mathbf{u} = \mathbf{0}, \mathbf{u} \neq \mathbf{0}$$

Thus:

- $\mathbf{u}$ is in the nullspace of $\lambda \mathbf{I} - \mathbf{A}$.
- $\det(\lambda \mathbf{I} - \mathbf{A}) = 0$

Equation $\det(\lambda \mathbf{I} - \mathbf{A}) = 0$ is called characteristic equation.

Characteristic Equation

- The order of characteristic equation is n .
- Characteristic equation has n roots, denoted $\lambda_1, \dots, \lambda_n$, possibly complex.
- $\mathbf{u}_i$ corresponding to λ_i can be easily found by finding the nullspace of $\lambda_i \mathbf{I} - \mathbf{A}$ matrix.

Eigenvalue and Eigenvector

Eigenvalue and Eigenvector

Find the eigenvalues and eigenvectors of $\mathbf{A} = \begin{bmatrix} 0.8 & 0.3 \\ 0.2 & 0.7 \end{bmatrix}$.

Solution:

$$\det(\mathbf{A} - \lambda \mathbf{I}) = \det \left(\begin{bmatrix} 0.8 - \lambda & 0.3 \\ 0.2 & 0.7 - \lambda \end{bmatrix} \right) = (\lambda - 1)(\lambda - 0.5) = 0$$

$$\Rightarrow \begin{cases} \lambda_1 = 1 \\ \lambda_2 = 0.5 \end{cases}$$

$$(\mathbf{A} - \lambda_1 \mathbf{I})\mathbf{u}_1 = \mathbf{0} \Rightarrow \begin{bmatrix} -0.2 & 0.3 \\ 0.2 & -0.3 \end{bmatrix} \mathbf{u}_1 = \mathbf{0} \Rightarrow \mathbf{u}_1 = \begin{bmatrix} 1.5 \\ 1 \end{bmatrix}$$

$$(\mathbf{A} - \lambda_2 \mathbf{I})\mathbf{u}_2 = \mathbf{0} \Rightarrow \begin{bmatrix} 0.3 & 0.3 \\ 0.2 & 0.2 \end{bmatrix} \mathbf{u}_2 = \mathbf{0} \Rightarrow \mathbf{u}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Eigenvalue and Eigenvector

Rank

The rank of matrix $\mathbf{A}$ is equal to the number of non-zero eigenvalues of $\mathbf{A}$.

Connection to Trace and Determinant

Assume $\mathbf{A} \in \mathbb{R}^{n \times n}$ with eigenvalues $\lambda_1, \dots, \lambda_n$. Then:

- The rank of $\mathbf{A}$ equals to the number of non-zero eigenvalues of $\mathbf{A}$.
- $\mathbf{A}^{-1}$ shares the eigenvector with $\mathbf{A}$ while its eigenvalues are $\frac{1}{\lambda_1}, \frac{1}{\lambda_2}, \dots, \frac{1}{\lambda_n}$.
- Symmetric matrix $\mathbf{A}$ is PD iff $\lambda_i > 0, i = 1, \dots, n$.
- Symmetric matrix $\mathbf{A}$ is PSD iff $\lambda_i \geq 0, i = 1, \dots, n$.
- $\text{tr}(\mathbf{A}) = \sum_{i=1}^n \lambda_i$
- $\det(\mathbf{A}) = \prod_{i=1}^n \lambda_i$

Diagonalizable

Diagonalizable

As we see: $A\mathbf{u}_i = \lambda_i\mathbf{u}_i, i = 1, \dots, n$

We can write the above equalities as:

$$AU = U\Lambda$$

where:

- $U \in \mathbb{R}^{n \times n} = \begin{bmatrix} | & | & | \\ \mathbf{u}_1 & \mathbf{u}_2 & \mathbf{u}_n \\ | & | & | \end{bmatrix}$
- $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$

Now assume that matrix U is invertible. Then:

$$A = U\Lambda U^{-1}$$

A matrix that can be written in this form is called diagonalizable.

Eigenvalues and Eigenvectors of Symmetric Matrices

Eigenvalues and Eigenvectors of Symmetric Matrices

Based on *Spectral Theorem*, for symmetric matrices we have:

- All eigenvalues are real
- Eigenvectors are orthonormal (U is orthogonal thus $U^{-1} = U^T$)

Then we have:

$$\begin{aligned} A = U \Lambda U^T &= \begin{bmatrix} | & | & | \\ \mathbf{u}_1 & \mathbf{u}_2 & \mathbf{u}_n \\ | & | & | \end{bmatrix} \begin{bmatrix} \lambda_1 & & \\ & \lambda_2 & \\ & & \ddots \\ & & & \lambda_n \end{bmatrix} \begin{bmatrix} - & \mathbf{u}_1^T & - \\ - & \mathbf{u}_2^T & - \\ & \vdots & \\ - & \mathbf{u}_m^T & - \end{bmatrix} \\ &= \sum_{i=1}^n \lambda_i \mathbf{u}_i \mathbf{u}_i^T \end{aligned}$$

Data Whitening Using Eigenvectors

Data Whitening Using Eigenvectors

Suppose we have a dataset $\mathbf{X} \in \mathbb{R}^{N \times D}$ where the empirical mean vector is zero and empirical covariance matrix is $\mathbf{\Sigma} = \frac{1}{N} \mathbf{X}^T \mathbf{X}$. Find matrix $\mathbf{W} \in \mathbb{R}^{D \times D}$ such that empirical covariance matrix for transformed vector $\mathbf{y} = \mathbf{W}\mathbf{x}$ is $\mathbf{I}$.

Solution: Matrix $\mathbf{\Sigma}$ is symmetric, thus $\mathbf{\Sigma} = \mathbf{U}\mathbf{D}\mathbf{U}^T$. Assume $\mathbf{W} = \mathbf{D}^{-\frac{1}{2}}\mathbf{U}^T$, then the covariance matrix for $\mathbf{y}$ is:

$$\begin{aligned}\text{Cov}[\mathbf{y}] &= \frac{1}{N} \mathbf{Y}^T \mathbf{Y} = \frac{1}{N} (\mathbf{X}\mathbf{W}^T)^T (\mathbf{X}\mathbf{W}^T) = \mathbf{W}\mathbf{\Sigma}\mathbf{W}^T \\ &= \mathbf{D}^{-\frac{1}{2}} \underbrace{\mathbf{U}^T \mathbf{U}}_{\mathbf{I}} \mathbf{D} \underbrace{\mathbf{U}^T \mathbf{U}}_{\mathbf{I}} \mathbf{D}^{-\frac{1}{2}} = \mathbf{D}^{-\frac{1}{2}} \mathbf{D} \mathbf{D}^{-\frac{1}{2}} = \mathbf{I}\end{aligned}$$

Section 9

Matrix Calculus

Gradient

Assume function $f : \mathbb{R}^n \rightarrow \mathbb{R}$. The gradient vector of this function at a point $\mathbf{x}$ is the vector of partial derivatives as:

$$\mathbf{g} = \frac{\partial f}{\partial \mathbf{x}} = \nabla f = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \\ \vdots \\ \frac{\partial f}{\partial x_n} \end{bmatrix}$$

To emphasize the gradient evaluation point we write:

$$\mathbf{g}(\mathbf{x}^*) \triangleq \frac{\partial f}{\partial \mathbf{x}} \Big|_{\mathbf{x}=\mathbf{x}^*}$$

Hessian

Assume function $f : \mathbb{R}^n \rightarrow \mathbb{R}$. The Hessian matrix of this function is the matrix of second partial derivatives as:

$$\mathbf{H}_f = \frac{\partial^2 f}{\partial \mathbf{x}^2} = \nabla^2 f = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \vdots & \cdots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1} & \cdots & \frac{\partial^2 f}{\partial x_n^2} \end{bmatrix}$$

Jacobian

Assume function $\mathbf{f} : \mathbb{R}^n \rightarrow \mathbb{R}^m$. The Jacobian matrix of this function is an $m \times n$ matrix of partial derivatives as:

$$\mathbf{J}_{\mathbf{f}}(\mathbf{x}) = \frac{\partial \mathbf{f}}{\partial \mathbf{x}^T} \triangleq \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1} & \cdots & \frac{\partial f_m}{\partial x_n} \end{bmatrix} = \begin{bmatrix} \nabla f_1(\mathbf{x})^T \\ \vdots \\ \nabla f_m(\mathbf{x})^T \end{bmatrix}$$